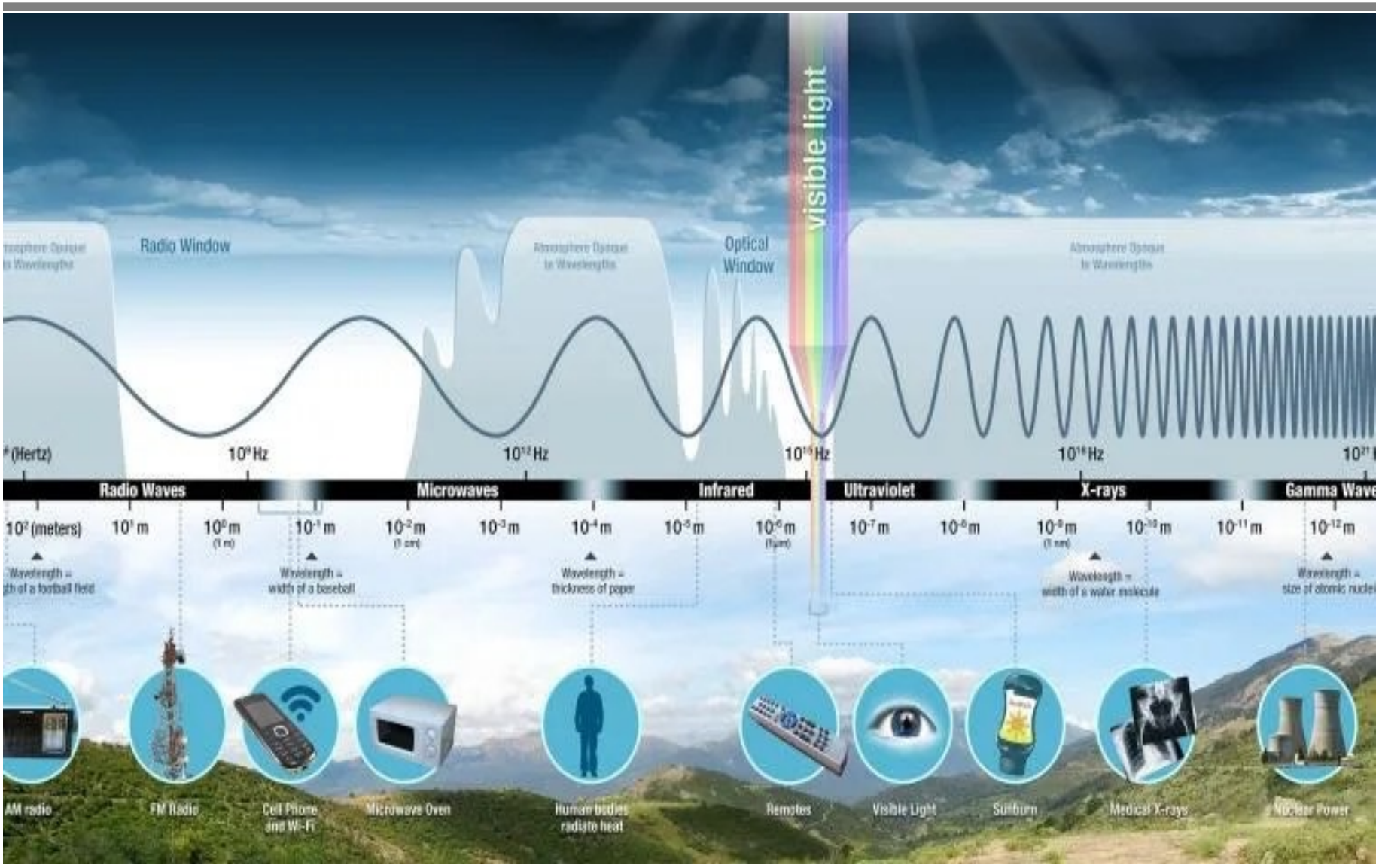
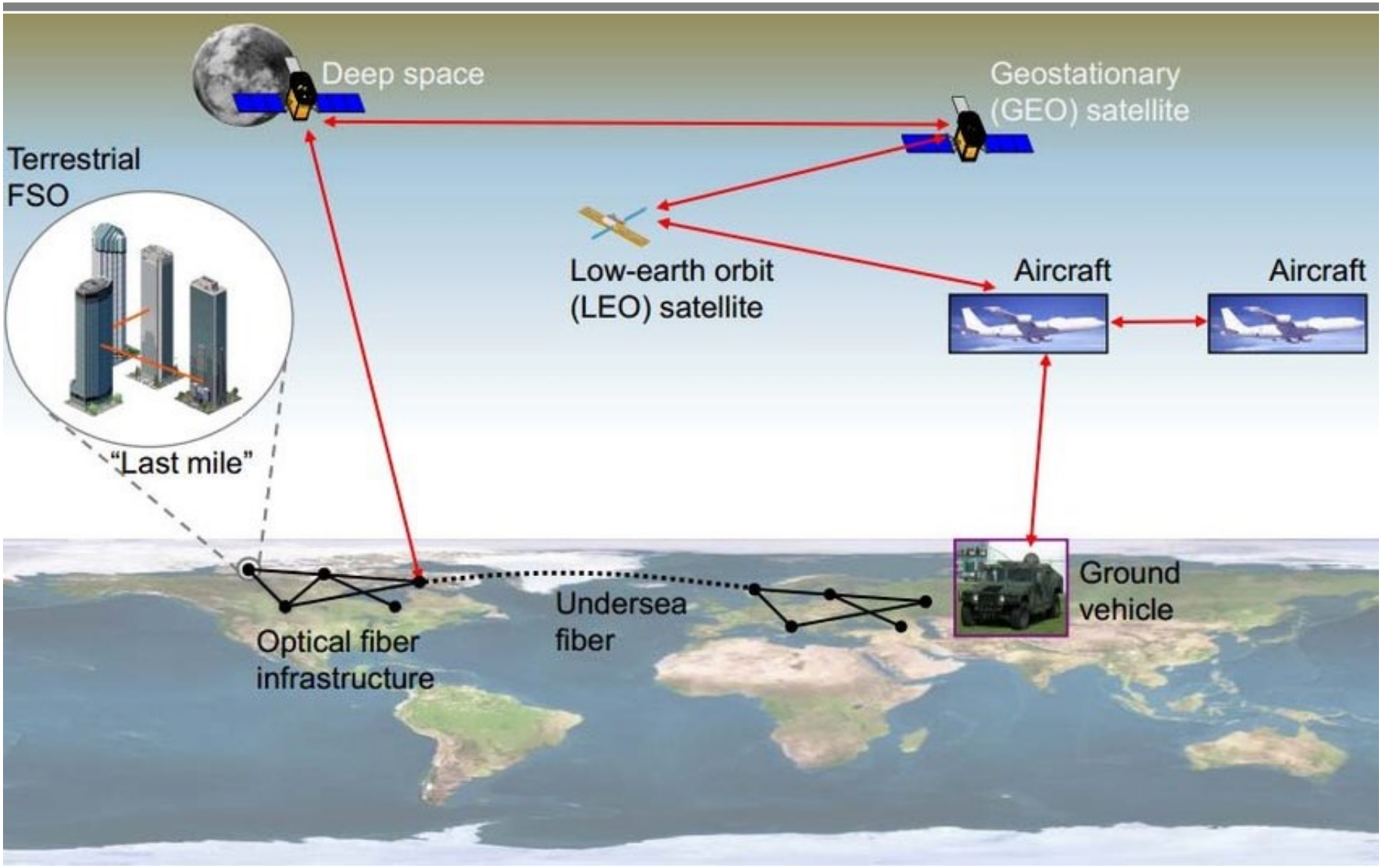
Introduction

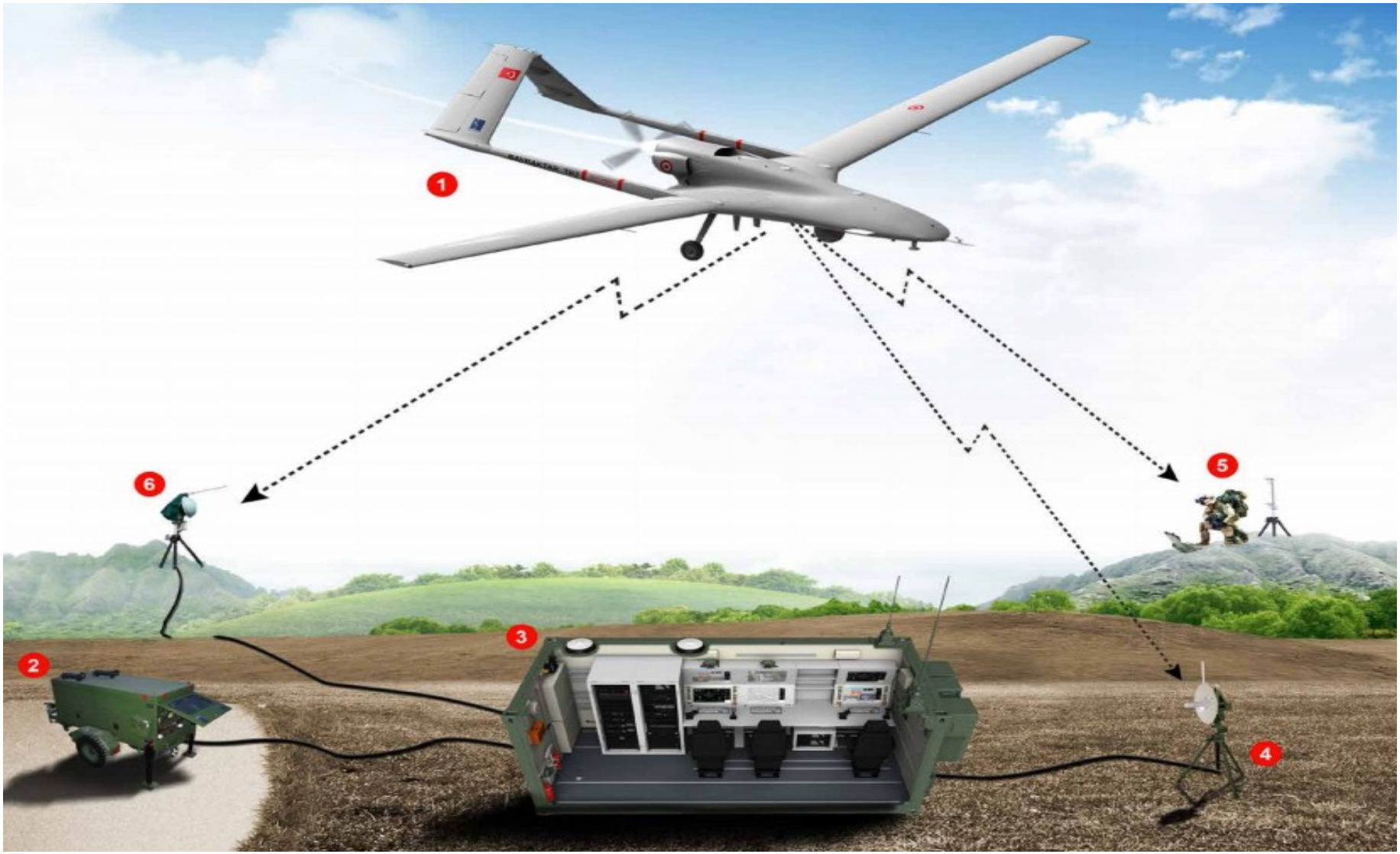
EM Waves - Spring 2024



EM Waves: Communications



EM Waves - Spring 2024



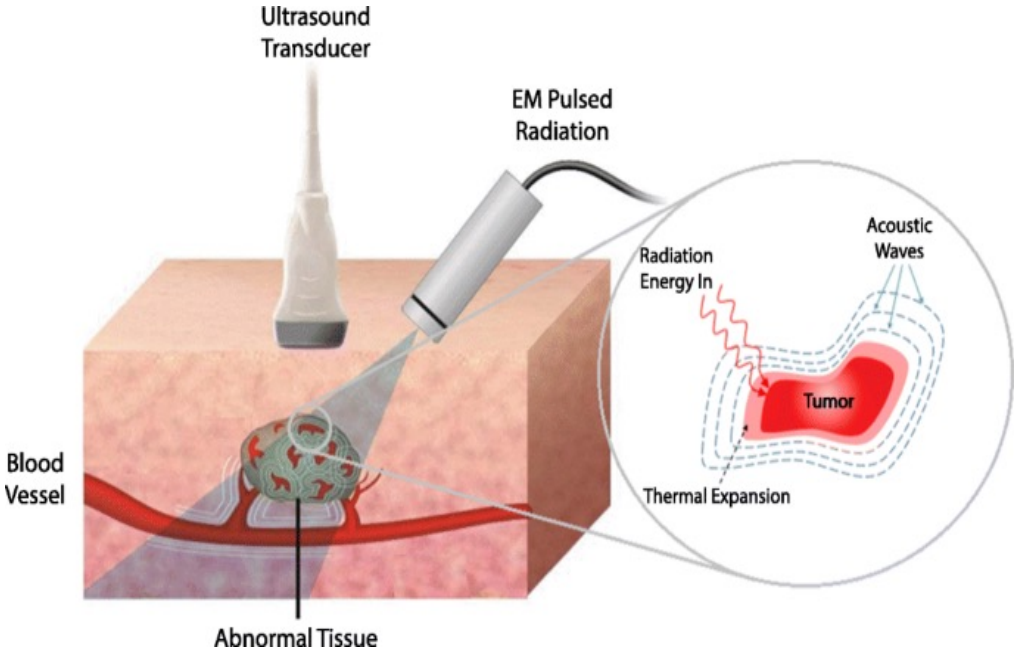
EM Waves: Remote Control

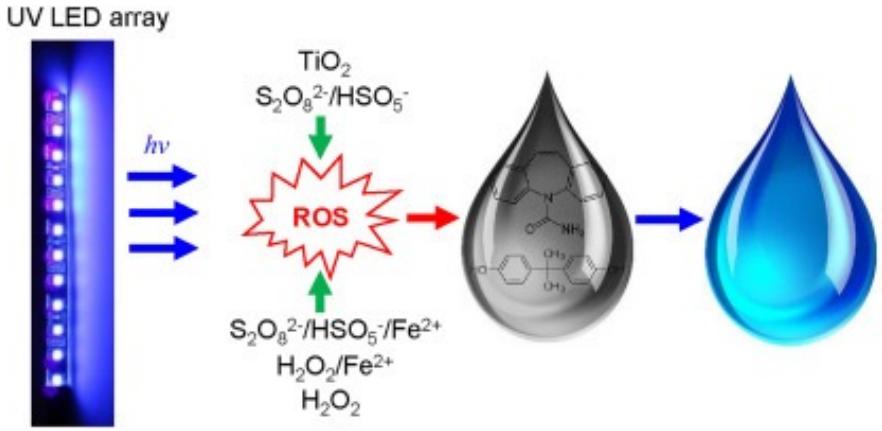


EM Waves: Sensing

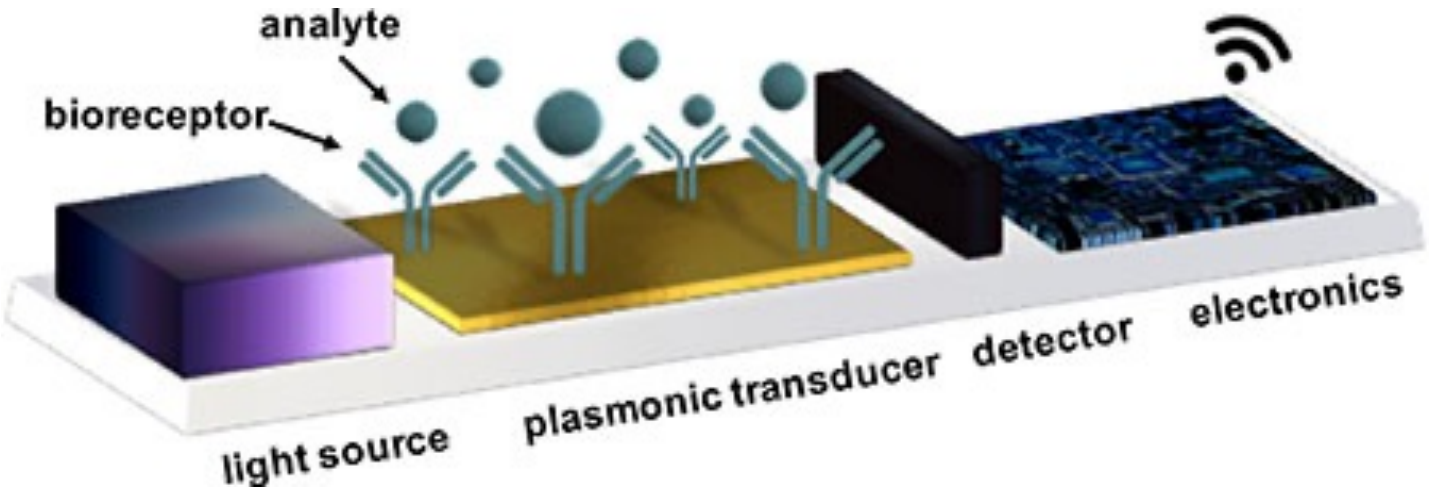


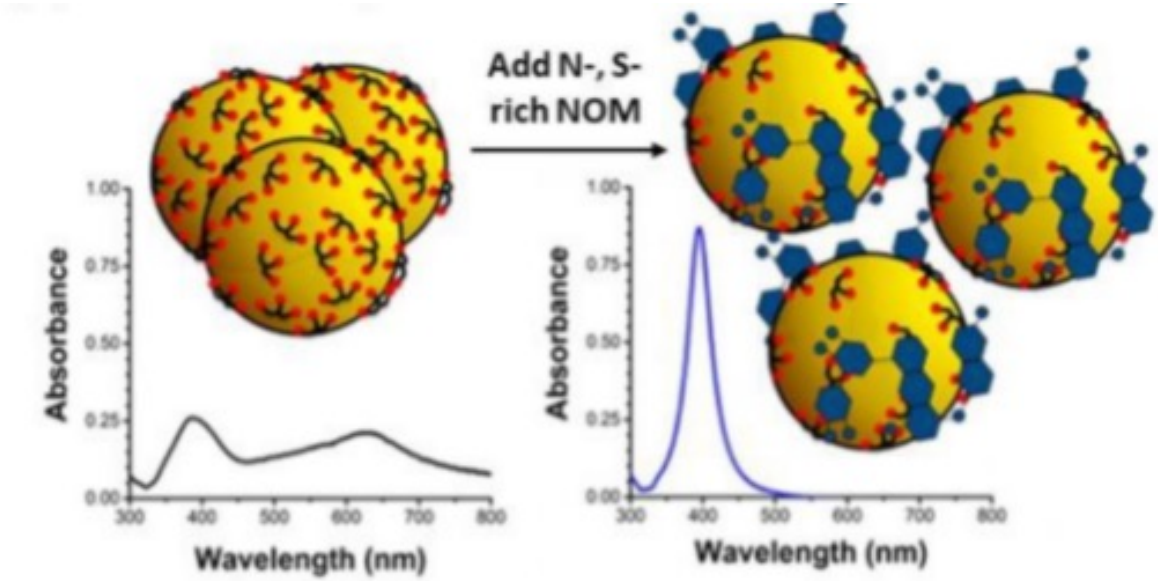






EM Waves: Bio-medical





EM Waves Subjects

The main subjects associated with the study of electromagnetic waves are

the Theory of EM Waves

the Generation of EM Waves

the Detection of EM Waves

the Management of EM Waves
(guiding, manipulation and amplification)

the Utilization of EM Waves
as a tool for the benefit of mankind

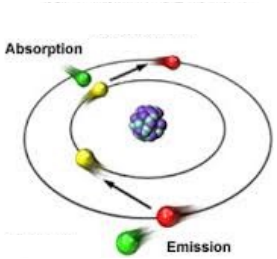
EM Waves Generation

Charge Oscillation: Dipoles





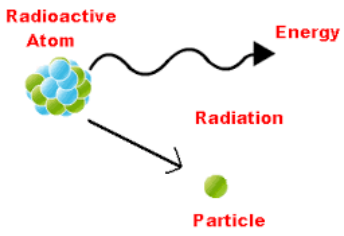

Electronic Transitions: EH Recombination







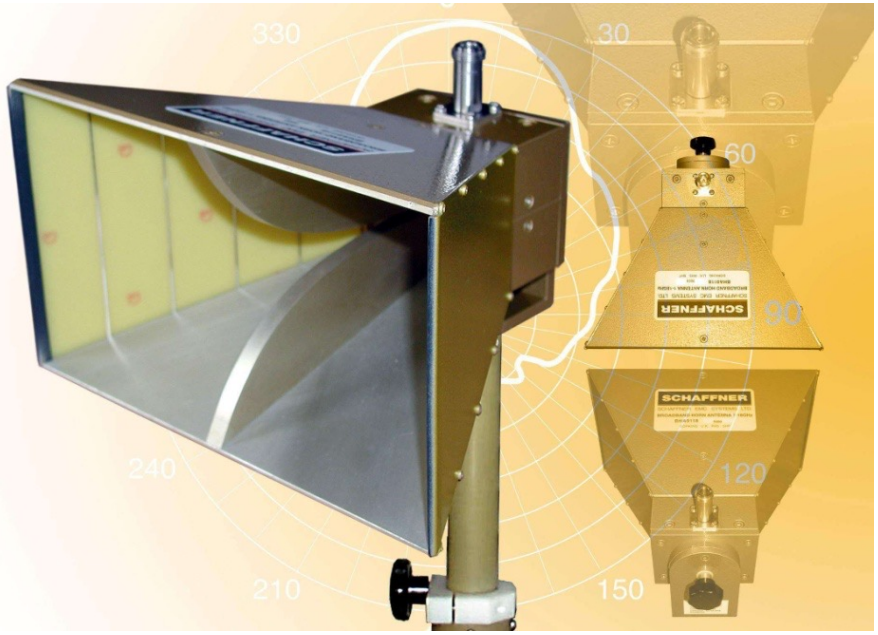
Atomic Disintegration: Decay

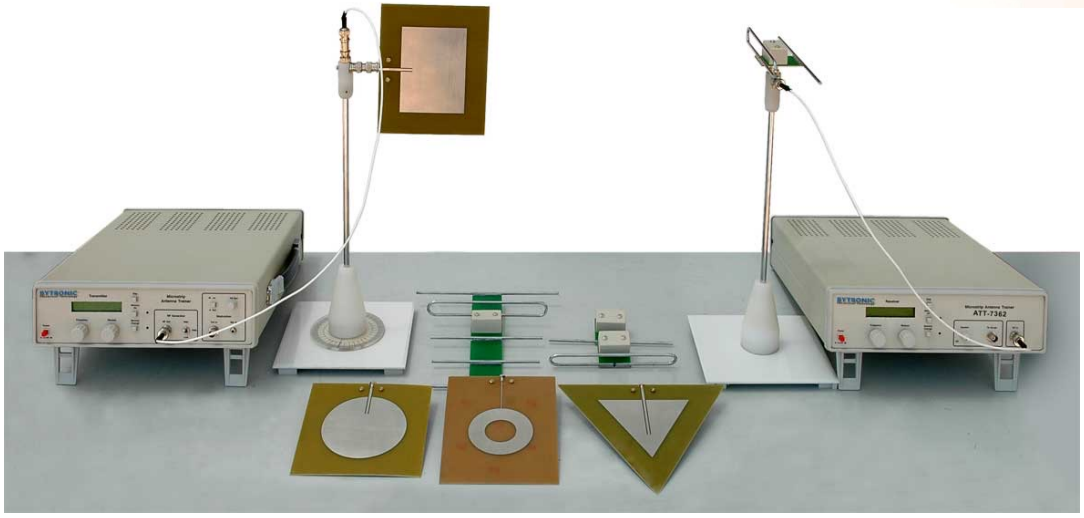




EM Waves Detection



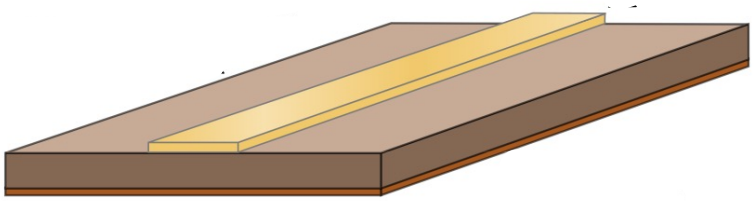




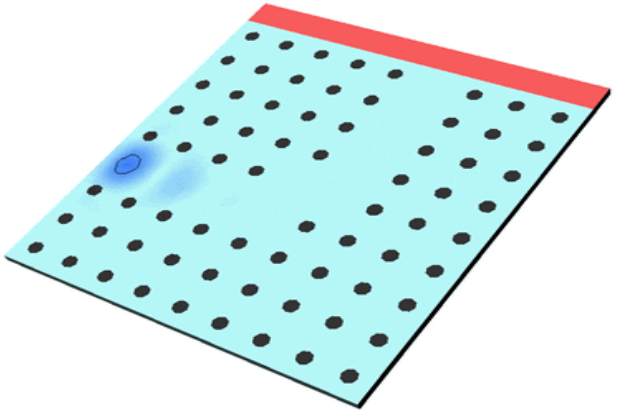


EM Waves Propagation: Guided





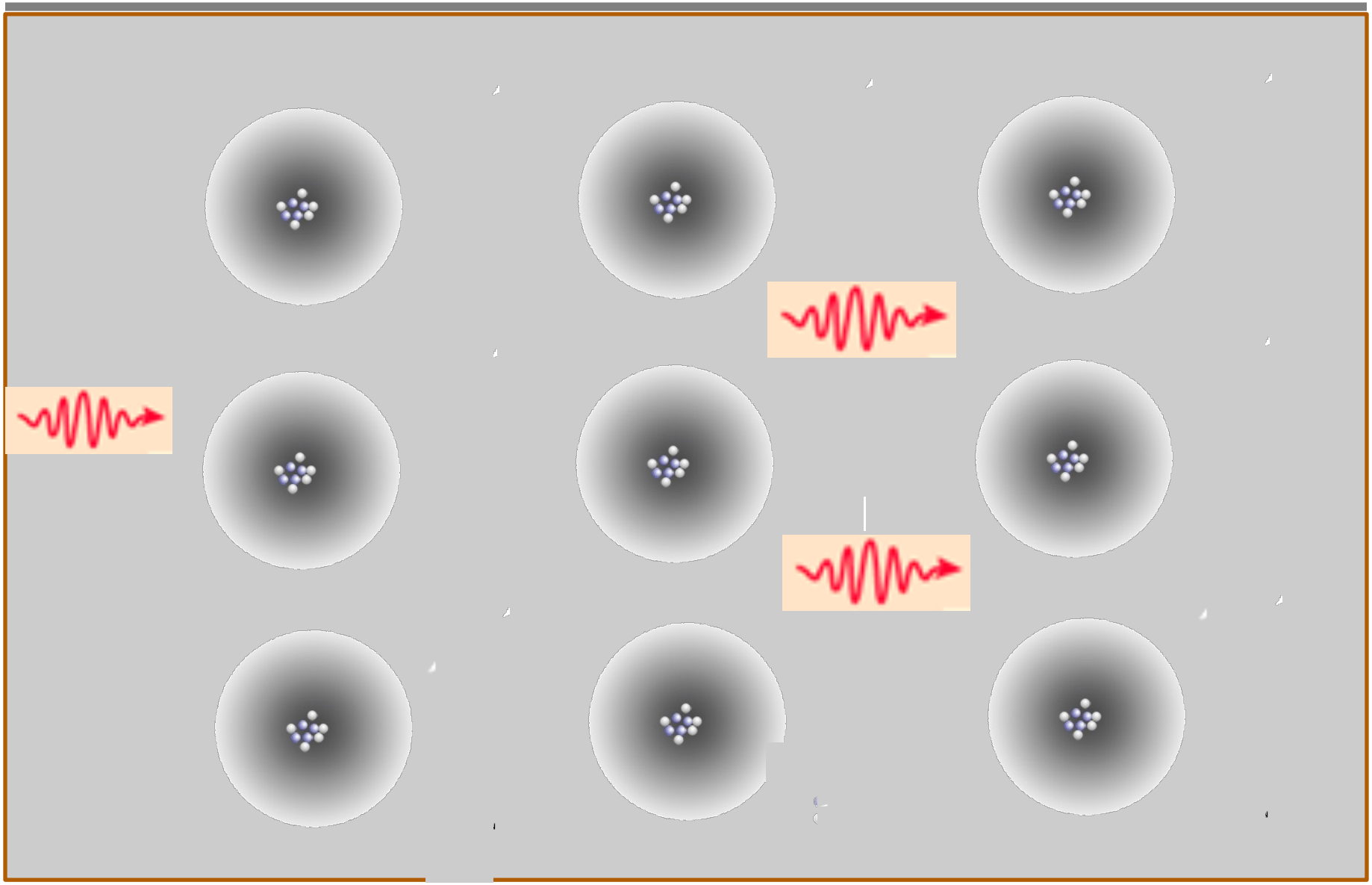




EM Waves Propagation: Guided



EM Wave-matter Interaction



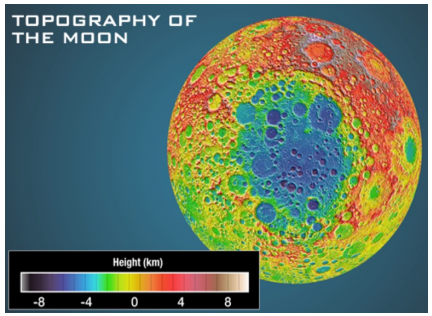
EM Wave-matter Interaction

Nano Photonics and Plasmonics Research

Reflection

Matter

Transmission

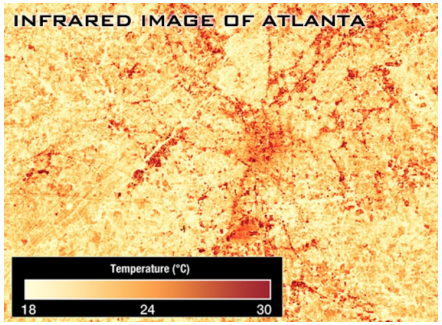




Refraction

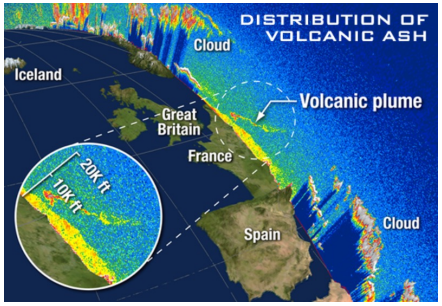
Propagation

Absorption



Radiation

Source



Scattering

EM Waves Subject is Multidisciplinary

Electrical Engineering:	The Electromagnetic wave nature. Microwave technology. Electronics and Optoelectronics. Communication Theory.
Physics/Chemistry:	Solid-state physics and Quantum mechanics.
Materials Science and Engineering:	Structure and Characterization. Crystallography. EOM and other properties. Synthesis and processing. Nanomaterials.
Others (on demand):	*for example*, Biomedical, Meteorology.

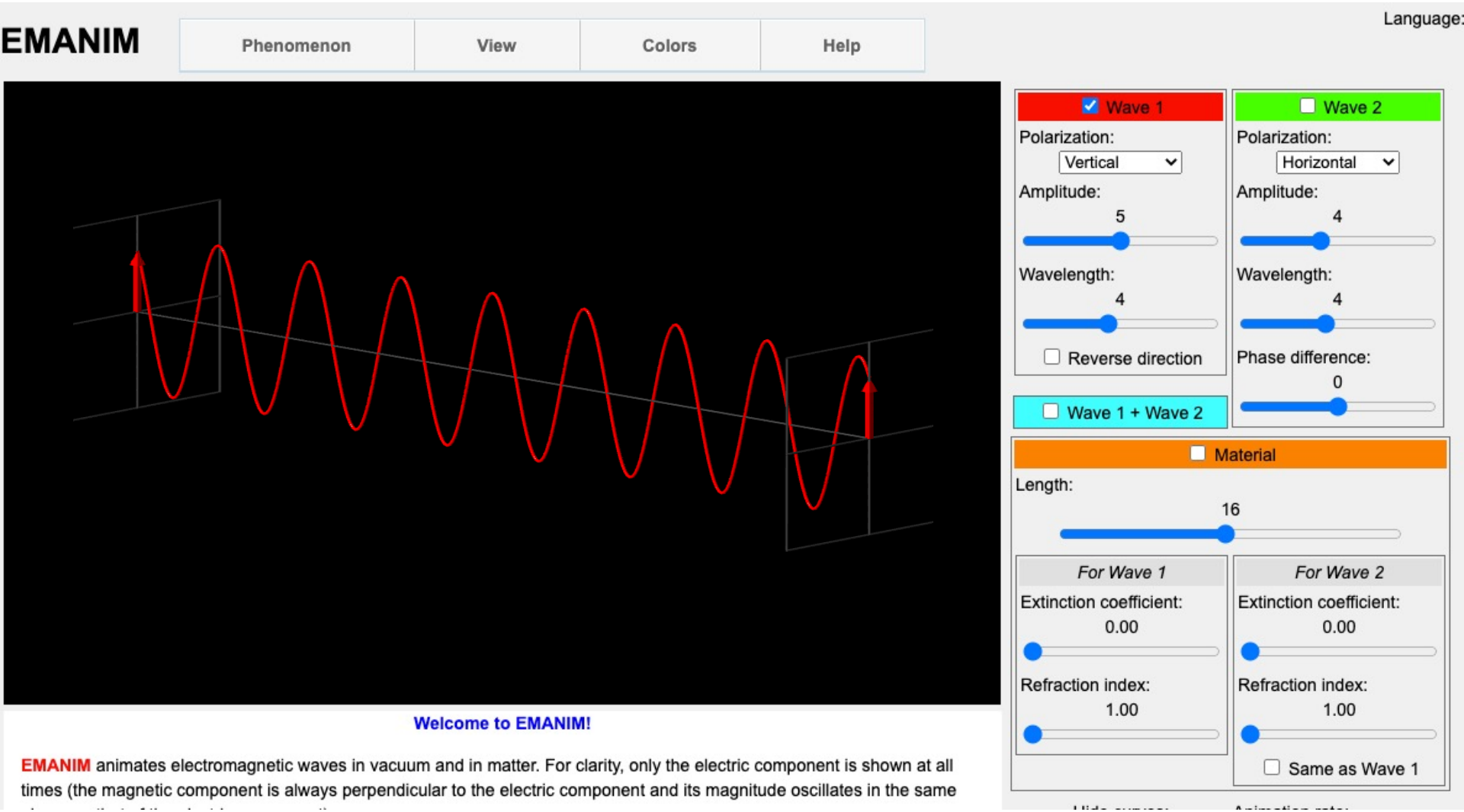
Syllabus

Nano Photonics and Plasmonics Research

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Phone	Office Hours MTWT: 13:00-14:00	Email: mohammad.alsunaidi@marmara	
Course Information	Text: M. Sadiku *Elements of Electromagnetics*	Grading	Class Work 30%		Midterm Exam 30%	Final Exam 40%

Week	Events	Subjects	Wednesdays	Fridays
1 Feb 12-16	---	Electromagnetic Waves: Theory and Analysis	Introduction to Electromagnetics.	Waves in General. Examples of EM Waves in One dimension
2 Feb 19-23	---		The Wave Equation and Solution, Types of Waves.	Representation of EM waves in 1D. Wave properties and Parameters.
3 Feb 26-Mar 1	Course drop/add Week		EM Spectrum, Wave-matter Interactions, Classification of Materials.	Maxwell's equations. Classification of EM Wave Problems.
4 Mar 4-8	---	Reflection, Refraction, Transmission and Absorption	Normal and Oblique Propagation.	TE and TM Representations.
5 Mar 11-15	Quiz 1 March 15 (In-class)		Single Interface at Normal Incidence.	Fresnel Equations. Examples.
6 Mar 18-22	---		Single Interface at Oblique Incidence.	Multiple Interfaces.
7 Mar 25-29	---		Interference and Anti Reflection Coating. Scattering	Wave-matter interaction. Absorption.
April 1-7	**Mid-term Exams period**			
Apr 9-12 Holidays				
9 Apr 15-19	---	EM Wave Guiding	Transmission Lines	Transmission Lines
10 Apr 22-26	Midterm Make-up (Excused, Week 11)		Transmission Lines	Transmission Lines
11 Apr 29-May 3	Labor Day: May 1		Waveguides	Waveguides
12 May 6-10	---		Cavities	Optical fibers
13 May 13-17	Quiz 2 May 17 (In-class)	EM Wave Generation	The Basic Hertzian Dipole Antenna.	Monopole and Half-Wave Dipole Antennas.
14 May 20-24	YS Day May 19		The Patch Antenna	Other Antenna Types
15 May 27-31	---	Special Topics	Topic 1	Topic 2
June 3-14	**Final Exams period**		**July 1-7 (Re-sit Final Exams period)**	

Animation and Visualization Tools



Maxwell's Equations: Where does the Wave Come From?

EM Fields Equations

Any electromagnetic field must satisfy these equations.

Differential Form

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{D} = \rho_v$$

$$\vec{\nabla}\cdot\vec{B} = 0$$

Integral Form

$$\oint \vec{E}\cdot\overrightarrow{dl} = \frac{d}{dt}\int \vec{B}(t)\cdot\overrightarrow{dS}$$

$$\oint \vec{H}\cdot\overrightarrow{dl} = \int \vec{J}\cdot\overrightarrow{dS} + \frac{d}{dt}\int \vec{D}(t)\cdot\overrightarrow{dS}$$

$$\oint \vec{D}\cdot\overrightarrow{dS} = Q$$

$$\oint \vec{B}\cdot\overrightarrow{dS} = 0$$

Continuity Equation

$$\vec{\nabla}\cdot\vec{J} + \frac{\partial}{\partial t}\rho_v = 0$$

EM Boundary Conditions

$$E_{1t} = E_{2t} \qquad D_{1n} = D_{2n} + \rho_s$$

$$\vec{H}_1\times\vec{a}_n - \vec{H}_2\times\vec{a}_n = \vec{K} \qquad B_{1n} = B_{2n}$$

Constitutive Relations (material)

$$\vec{J} = \sigma\vec{E} \qquad \vec{B} = \mu\vec{H} \qquad \vec{D} = \varepsilon\vec{E}$$

EM Energy Density

$$w_E = \frac{\varepsilon}{2}\left|\vec{E}\right|^2 \qquad w_m = \frac{\varepsilon}{2}\left|\vec{H}\right|^2$$

The Wave Equation

The wave equation is a general differential equation that gives the mathematical description of the travel of a wave.

The derivation of this equation follows from the dynamics of the situation. For example, the wave equation associated with a stretched string can be obtained from the equations of motion (Newton's laws).

The simplified general scalar form in 1D

$$\frac{\partial^2 \varphi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \varphi}{\partial t^2}$$ (Rectangular coordinates)

In 3D (scalar)

$$\nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 \varphi}{\partial t^2}$$

The simplified general vector form in 3D

$$\vec{\nabla}^2 \vec{E} = \frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 \vec{E}}{\partial t^2}$$

Maxwell's Equations

Maxwell's Equations

$$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{D} = \rho_v$$

$$\vec{\nabla}\cdot\vec{B} = 0$$

Constitutive Relations (material)

$$\vec{J} = \sigma\vec{E} \qquad \vec{B} = \mu\vec{H}$$

$$\vec{D} = \varepsilon\vec{E}$$

Starting with $$\vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu\frac{\partial \vec{H}}{\partial t}$$

Take the curl of both sides

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = -\mu\frac{\partial\,(\vec{\nabla}\times\vec{H})}{\partial t}$$

Use the vector identity

$$\vec{\nabla}\times\vec{\nabla}\times\vec{E} = \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2\vec{E}$$

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2\vec{E} = -\mu\frac{\partial}{\partial t}\left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right)$$

$$\vec{\nabla}^2\vec{E} - \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

If the medium is charge-free, $\vec{\nabla}\cdot\vec{D} = 0 \rightarrow \vec{\nabla}\cdot\vec{E} = 0$

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial \vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial \vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

The EM Wave Equation

The EM Wave Equation

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Write the wave equation in this form

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}$$

Wave speed $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ For free space $v = \dfrac{1}{\sqrt{\mu_o\varepsilon_o}} = 3.0\times10^8$ (m/s)

1) Electromagnetic waves exist.

2) The wave is composed of two interacting quantities E and H fields.

3) EM waves travel with the speed of light.

4) Light is an EM wave.

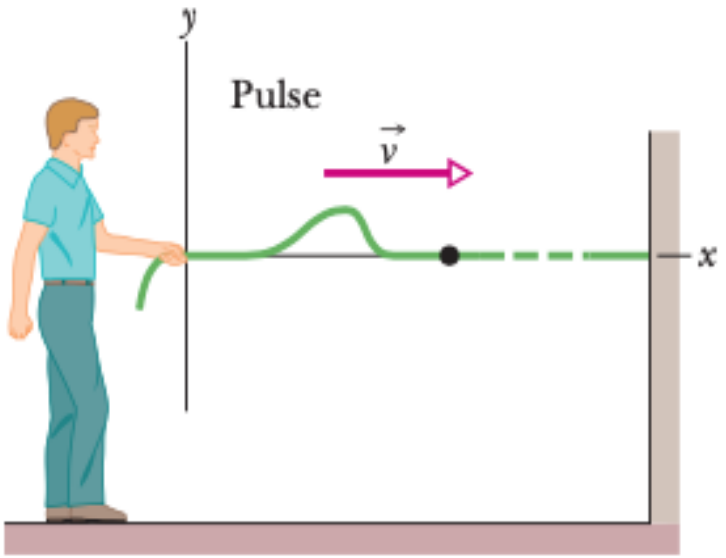
Waves in General

The following subjects will be discussed

- Types of Waves.
- The Wave Equation.
- SHM and Time-harmonic Waves.
- Wave parameters: Speed, Frequency and Wavelength.
- Forced Oscillations and Resonance.

What is a Wave?

- What we call a wave is a disturbance that travels through a certain medium.
- The velocity of the wave is determined by the medium.
- The wave and the medium are two different things.
- In this view, the medium is subjected to the effect of the wave.
- The wave is like a ghost; only the effect of the ghost is observed.





Types of Waves

Waves can be classified according to their sources, as

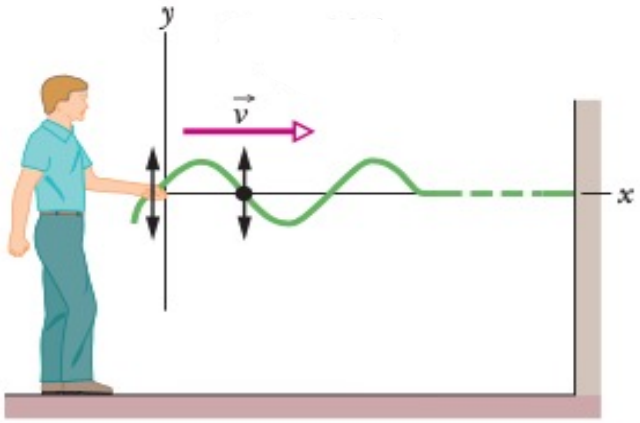
1) Mechanical Waves: Waves associated with the vibration of particles.

2) Electromagnetic Waves: Waves associated with the acceleration of electric charges

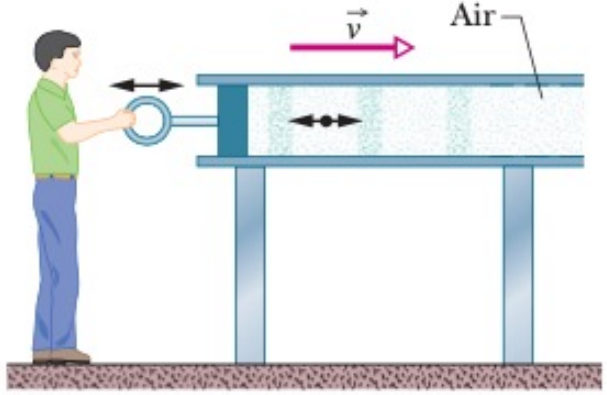
3) Matter Waves: Waves associated with the matter elementary particles.

Waves can be classified according to their behavior, as

1) Transverse Waves: Medium "particles" oscillate normal to the direction of propagation of the wave.



2) Longitudinal Waves: Medium "particles" oscillate in the same direction as the wave propagation direction.



The Wave Equation

The wave equation is a general differential equation that gives the mathematical description of the travel of a wave.

The derivation of this equation follows from the dynamics of the situation. For example, the wave equation associated with a stretched string can be obtained from the equations of motion (Newton's laws).

The derivation of EM wave equation is obtained from the manipulation of Maxwell's equations.

The simplified general scalar form in 1D

$$\frac{\partial^2 \varphi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 \varphi}{\partial t^2}$$ (Rectangular coordinates)

In 3D (scalar)

$$\nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 \varphi}{\partial t^2}$$

The simplified general vector form in 3D

$$\vec{\nabla}^2 \vec{E} = \frac{\partial^2 \vec{E}}{\partial x^2} + \frac{\partial^2 \vec{E}}{\partial y^2} + \frac{\partial^2 \vec{E}}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 \vec{E}}{\partial t^2}$$

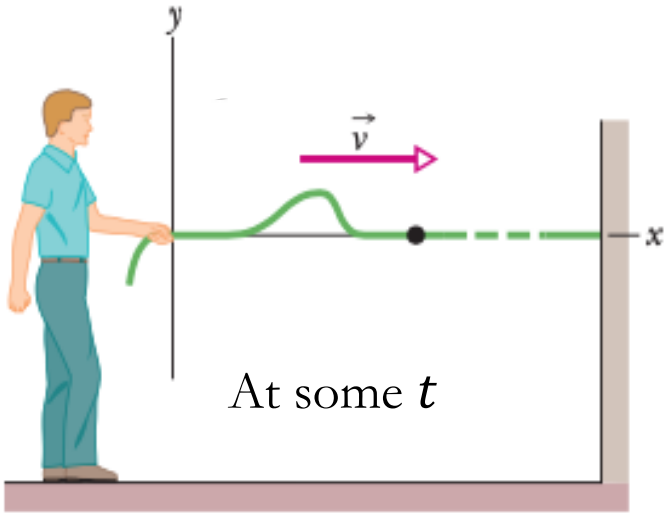
The Wave Equation

Let's stay with the 1D scalar wave equation. $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$

This equation can express, for example, wave propagation in a stretched string along the x axis.

In this case, $y(x,t)$ represents the transverse displacement of the elements of the string at any time t and at any location x.

The wave, as a whole, travels with an energy velocity v.



In the case of EM waves, what does this displacement function represent?

What are possible solutions of this differential equation?

The Sinusoidal Solution

Let's stay with the 1D scalar wave equation.

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

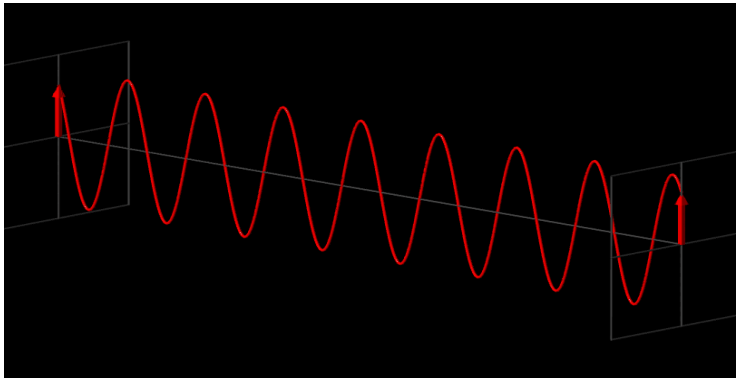
Possible Solution

The displacement that happens in the medium at any time and location is given by,

Amplitude — Phase

$$y(x,t) = A\ \sin(\omega t - \beta x)$$

Frequency — Wave (phase) number



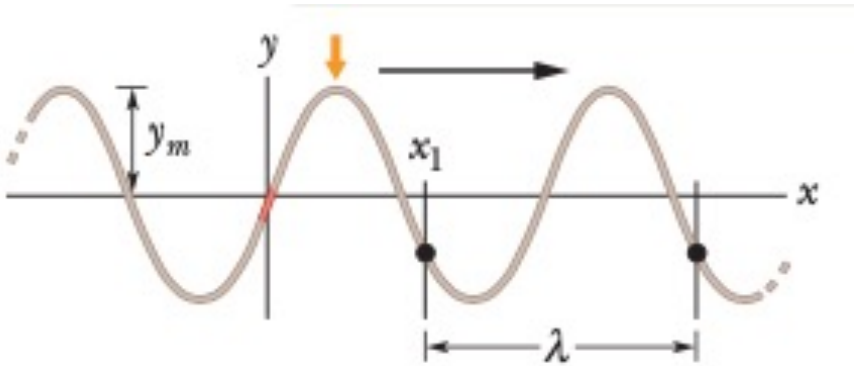
$$v = \frac{\omega}{\beta}$$

Wave velocity

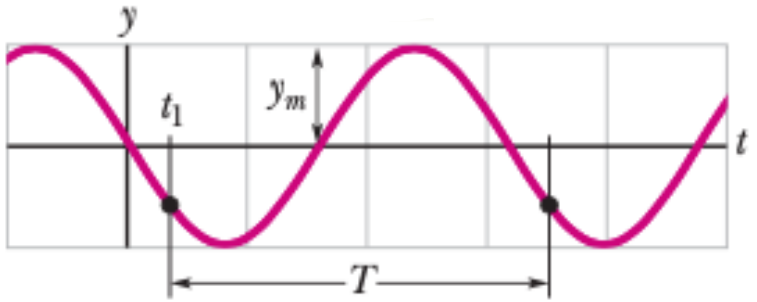
The Sinusoidal Solution

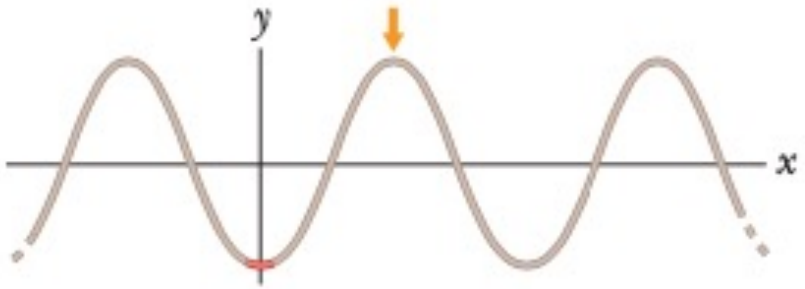
$$y(x,t) = A\ \sin(\omega t - \beta x)$$

A snapshot

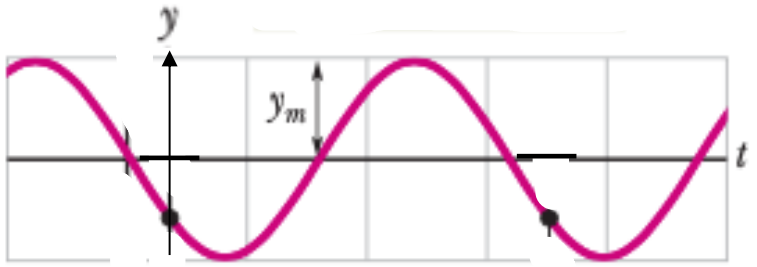


One location





Later in time



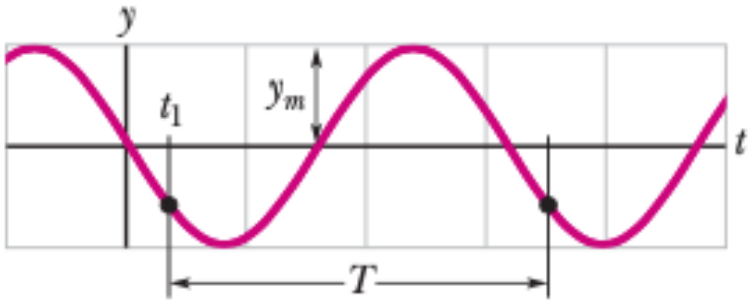
Another location

The Sinusoidal Solution: Frequency

$$y(x,t) = A\ \sin(\omega t - \beta x)$$

The frequency ω is the number of temporal cycles per one second (rad/s).

Take $x = 0$ for simplicity, for any two points separated by time of T,



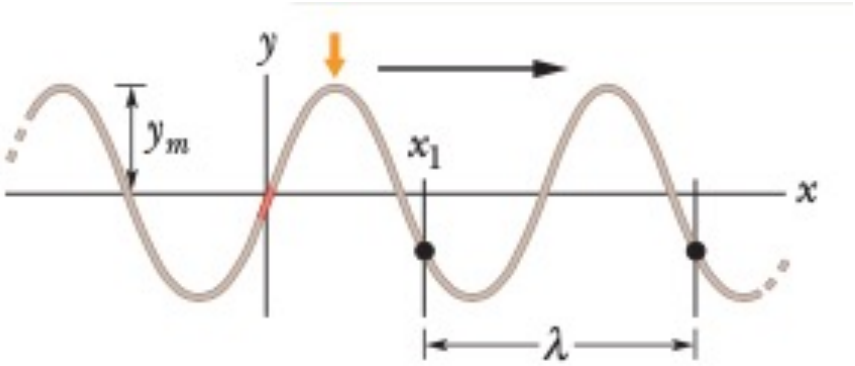
$$\sin(\omega t) = \sin(\omega(t+T)) = \sin(\omega t + \omega T) \quad \Rightarrow \quad \omega = \frac{2\pi}{T} \quad \text{(rad/s)}$$

The Sinusoidal Solution: Wave Number

$$y(x,t) = A\ \sin(\omega t - \beta x)$$

The phase (wave) number β is the "spatial frequency", or the number of spatial cycles per one meter (rad/m).

Take $t = 0$ for simplicity, for any two points separated by distance of λ,

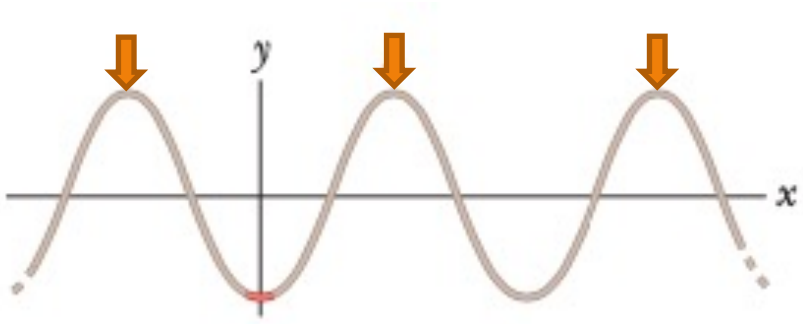


$$\sin(\beta x) = \sin(\beta(x+\lambda)) = \sin(\beta x + \beta\lambda) \quad \Rightarrow \quad \beta = \frac{2\pi}{\lambda} \quad \text{(rad/m)}$$

The Sinusoidal Solution: Velocity

$$y(x,t) = A\ \sin(\omega t - \beta x)$$

In order for the wave to keep the same form, the argument of the sinusoid has to be constant for similar points.



$$\omega t - \beta x = constant \quad \Rightarrow \quad \omega - \beta \frac{\partial x}{\partial t} = 0 \quad \Rightarrow \quad \omega = \beta v$$

Phase velocity $v = \frac{\omega}{\beta}$ (m/s)

The velocity is positive; the wave is travelling in the positive x direction.

Examples

1) Consider a wave described by $F(x,t) = 50\cos(10^8 t + \frac{x}{3})$ (a.u.)

Find the direction of propagation, speed and wavelength.

The direction of propagation, $-x$ direction.

Frequency $\omega = 10^8$ (rad/s)

Phase constant (wavenumber) $\beta = \frac{1}{3}$ (rad/m)

Speed (phase velocity) $v = \frac{\omega}{\beta} = \frac{10^8}{1/3} = 3\times10^8$ (m/s)

Wavelength $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{1/3} = 6\pi$ (m)

Examples

2) Consider a wave described by $S(z,t) = 2\sin(12t - \beta z + \frac{\pi}{3})$ (a.u.)

The wave speed is known to be 330 m/s. Find the direction of propagation, phase constant, wavelength and wave amplitude at $t = 2.6$ s and $z = 5.5$ m.

The direction of propagation, $+z$ direction.

Frequency $\omega = 12$ (rad/s)

Phase constant (wavenumber) $\beta = \frac{\omega}{v} = \frac{12}{330}$ (rad/m)

Wavelength $\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{12/330} = \frac{660\pi}{12}$ (m)

Amplitude $S(5.5, 2.6) = 1.18$ (a.u.)

Q & A

Send me an email if you have any questions.